ATAC-seq Pipeline

Version 2.0.0

Overview

Samples were processed with the BIC ATAC-seq pipeline, which takes aligned, duplicate-marked BAM files and produces filtered alignments, normalized signal tracks (bigWig), MACS2 peak calls for each sample, a merged peak atlas with a raw fragment count matrix, QC metrics and, when the design has replicates, a differential accessibility analysis. The code is at https://github.com/soccin/ATAC-seq/releases/tag/v2.0.0.

Read filtering, the Tn5 shift and peak calling follow the ENCODE ATAC-seq pipeline (ENCODE-DCC/atac-seq-pipeline v2.2.3). The signal tracks, the peak atlas and the count normalization follow the methods of R. Koche (MSKCC Center for Epigenetics Research).

Reads were aligned to one of three reference genomes: GRCh37 (b37), GRCh38 (the GDC `GRCh38.d1.vd1` build, hg38) or GRCm38 (mm10). The build of a project is recorded in the BAM headers. Peaks, signal tracks and the atlas use the autosomes, X and Y only.

Methods

Alignment. Paired-end reads were aligned with BWA-MEM 0.7.19 (Li 2013) using the PEMapper pipeline (https://github.com/soccin/PEMapper), which trims adapters with cutadapt, aligns with `bwa mem -M`, merges lanes, sorts, and marks duplicates with Picard MarkDuplicates 2.25.5. Duplicates are marked in the BAM, not removed. The PEMapper version is in the `@PG` header line of each BAM.

Read filtering. With samtools 1.20 (Danecek 2021) each BAM was reduced to properly paired reads with mapping quality 10 or more, without unmapped reads, reads whose mate is unmapped, secondary alignments, reads failing vendor QC or duplicates (`-q 10 -f 3 -F 1804`). Two further filters, which the ENCODE pipeline does not have, correct for an artifact of BWA-MEM local alignment. A read whose sequence is not from the reference, or only partly from it, can still be reported as a short alignment of 19 to 35 bp with the rest of the read soft-clipped. Such reads carry the same non-reference sequence, so they pile up at single positions in a few hundred loci, and in some libraries they are half of all read pairs. Bowtie2 in end-to-end mode, the ENCODE aligner, cannot produce these alignments, so ENCODE needs no such filter. The first filter removes every read whose alignment is soft-clipped at its 5' end, the Tn5 insertion site: a forward-strand read whose CIGAR string starts with `S` or a reverse-strand read whose CIGAR string ends with `S`. The reads were then name-sorted and passed through `samtools fixmate -r`, and the flag filter was applied again, as in the ENCODE pipeline, which removes a read whose mate did not pass the filters above. The second filter was applied at the same step: it removes every pair with an insert of 30 bp or less (|TLEN| <= 30). The result was coordinate-sorted with Picard SortSam 3.4.0; this filtered BAM is the input to every later step.

Tn5 shift. Each read was converted to a BED interval with bedtools 2.31.1 (Quinlan 2010) and shifted to place its 5' end at the Tn5 insertion site: +4 bp on the forward strand and -5 bp on the reverse strand (Buenrostro 2013). Reads on the mitochondrial contig and on unplaced, random, decoy and viral contigs were dropped at this step, as in the ENCODE pipeline; the fraction of mitochondrial reads is reported as a QC metric. This set of shifted reads is the input to the signal

tracks and to peak calling.

Signal tracks. For each sample the coverage of the shifted reads was computed with bedtools genomecov, each read as its own interval with no extension, scaled to 10 million reads (the scale factor is 10 million divided by the number of shifted reads of the sample), and written as a bigWig with the UCSC wigToBigWig tool (Kent 2010). Tracks are therefore comparable between samples by sequencing depth; the DESeq2 scale factors below are not applied to them.

Peak calling. Peaks were called for each sample with MACS2 2.2.9.1 (Zhang 2008) on the shifted reads, with the settings of the ENCODE ATAC-seq pipeline: `--nomodel --shift -75 --extsize 150 --keep-dup all --call-summits -p 0.01` and the genome size `hs` or `mm`. Each read contributes a 150 bp window centered on its insertion site, every read is kept (duplicates were already removed by the filter), and peaks are called at a p-value of 0.01, a permissive cutoff chosen by ENCODE so that peaks can be ranked downstream. MACS2's own output (narrowPeak, summits, xls) is delivered unchanged. The peaks were then post-processed as ENCODE does: sorted by p-value, the 300,000 most significant kept, clipped to the contig ends, peaks overlapping the ENCODE blacklist of the genome removed (Amemiya 2019; Boyle lab blacklist v2 for hg38 and mm10, the DAC consensus excludable list for b37) and the score column capped at 1000. These `blacklistFiltered` peaks and summits are delivered beside MACS2's files and are the peaks used for the atlas.

Peak atlas. The blacklist-filtered peaks of all samples were pooled and merged with bedtools merge, joining peaks within 500 bp of each other into one region (R. Koche's method; ENCODE does not merge peaks). Any blacklisted region bridged by a merge was subtracted, so no atlas region overlaps the blacklist. The result is the peak atlas, one set of regions shared by every sample of the project, written in SAF format (1-based, inclusive coordinates) with the regions named `Peak_1`, `Peak_2`, … in genome order. Because peaks are merged and pooled over samples, the number of atlas regions grows with the number of samples and is not comparable to ENCODE's per-sample peak-count standards; the per-sample `narrowPeak` files are.

Counts. Fragments were counted over the atlas with featureCounts 2.1.1 (Liao 2014) from the filtered BAMs, counting read pairs as single fragments (`-p --countReadPairs`), with a fragment that overlaps two atlas regions counted in both (`-O`), and with the same mapping quality cutoff as the read filter (`-Q 10`). The counts were verified by two independent checks before use: each sample's fragment total must be half its mapped reads, within 5%, and the number of fragments in the atlas, counted a second time with samtools alone, must equal featureCounts' assigned total exactly. A run whose counts fail either check is stopped.

Normalization. Size factors were computed from the raw count matrix with DESeq2 (Love 2014; median-of-ratios method, Anders 2010). They are reported as scale factors, the reciprocal of the DESeq2 size factor, so that a normalized count is the raw count multiplied by the sample's scale factor.

QC. The insert-size distribution of the filtered BAM was computed with Picard CollectInsertSizeMetrics over a fixed range of 0 to 1000 bp, as in the ENCODE pipeline, and plotted for all samples, to show the nucleosome-free and nucleosomal fragment populations. Its smallest insert is 31 bp, because of the insert filter above. The mitochondrial fraction is the number of reads on the mitochondrial contig divided by all reads of the filtered BAM, so it is computed after duplicate removal; ENCODE computes its fraction on all alignments before filtering, and because mitochondrial reads are highly duplicated the fraction here is the lower of the two. TSS enrichment was computed with an R port of ENCODE's `encode_task_tss_enrich.py`: the coverage of the shifted reads in a window

of 2 kb on each side of every unique protein-coding TSS, averaged over TSSs in 400 bins and divided by the mean coverage of the outer 100 bp on each side; the score is the maximum of this profile. The TSS sets are the GENCODE v24 (hg38), v19 (b37) and vM7 (mm10) sets of the ENCODE pipeline; ENCODE's cutoffs for the score, given with the output, were set on these annotations and are a guide. The fraction of fragments in peaks is the number of fragments featureCounts assigned to the atlas divided by all fragments of the filtered BAM, taken from the featureCounts summary. It measures the same thing as ENCODE's FRiP, signal against background, but is a different quantity: the regions are the merged atlas of all samples rather than the sample's own peaks, and the denominator includes mitochondrial fragments, so it is not comparable to ENCODE's FRiP cutoffs. It and a principal component analysis of the samples (edgeR TMM-normalized log2 CPM over the atlas, after filterByExpr) were plotted from the count matrix.

Differential accessibility. When the design has replicates, pairwise comparisons between groups were tested with edgeR (Robinson 2010; McCarthy 2012). For each comparison, the samples of its two groups were taken from the count matrix, atlas regions with too few counts removed with filterByExpr, library sizes normalized with TMM, dispersions estimated, and a negative binomial generalized linear model fitted and tested with the likelihood ratio test. P-values were adjusted with the Benjamini-Hochberg method (Benjamini 1995) and regions with FDR < 0.05 were called significant. The sign of the log fold change is group 2 minus group 1 of the comparison: a positive value means more accessible in group 2. Significant regions were annotated with the nearest gene and genomic feature using ChIPseeker (Yu 2015) with the UCSC knownGene annotation of the build and a TSS region of 5 kb on each side.

Tool versions

Tool	Version
BWA-MEM (PEMapper)	0.7.19
Picard MarkDuplicates (PEMapper)	2.25.5
samtools	1.20
Picard SortSam, CollectInsertSizeMetrics	3.4.0
bedtools	2.31.1
UCSC wigToBigWig, bedClip	v369
MACS2	2.2.9.1
featureCounts (subread)	2.1.1
R	4.5.1
ENCODE ATAC-seq pipeline (settings followed)	2.2.3

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